Activity 05 - Use AI Analysis to Build a Native PowerPoint Data Story

- Course: Infographics and Data Visualization with PowerPoint (TGS-2026064178)
- Duration: 45 minutes
- Tools: Approved AI assistant + Excel + native PowerPoint chart
- Output: Formula-driven workbook, native editable chart and annotated decision slide

Objective

Use ai to clean and analyse mock excel data, verify calculations, then create a native editable chart and decision story in powerpoint.

Scenario

Northstar's marketing team must choose one product family for the next promotion. The supplied workbook contains monthly revenue, units, margin and returns for four categories. Prompt an approved AI assistant to analyse the workbook and propose candidate insights. Reproduce the selected calculations in Excel, then create an editable PowerPoint chart with an explicit decision implication.

Detailed instructions

1. Open and validate the workbook. Open the A05 XLSX. Review Data, Analysis, Source Notes and AI Prompt Lab. Change one input value and confirm formulas update without errors.

2. Run the analysis prompt. Ask an approved AI assistant to flag data-quality issues, compare revenue, growth, margin and returns, calculate candidate metrics and provide citations to the relevant rows. Require no invented values.

3. Verify every calculation. Reproduce the selected growth, totals, averages and comparisons with Excel formulas. Record any rejected AI claim and why.

4. Define the decision metric. State which product family to promote and the minimum evidence required. Separate observed patterns from hypotheses and caveats.

5. Choose the chart. Use a clustered bar for category comparison or a line for change over time. Explain why the chart reveals the relationship better than a pie or 3-D chart.

6. Create the Excel chart. Select the prepared Analysis range, choose Insert > Recommended Charts, select the intended chart, and add a precise chart title.

7. Paste as a native chart. Copy the chart. In PowerPoint choose Home > Paste > Use Destination Theme & Embed Workbook or Keep Source Formatting & Link Data. Do not paste as a picture.

8. Remove chart junk. Use Chart Design > Add Chart Element. Keep only necessary axes, direct labels and a short title. Remove background fills, 3-D effects, unnecessary gridlines and legends.

9. Write the action title. Replace the topic title with a sentence that states the supported conclusion, for example Ready Meals combine the fastest Q2 growth with above-floor margin.

10. Highlight the decision. Use one accent colour for the selected category. Keep all comparison categories neutral and add one concise annotation.

11. Add source and limitation. Show Synthetic Northstar data, Jan-Jun 2026 and one relevant caveat such as promotional seasonality or return-rate uncertainty.

12. Test refresh and log. Change one workbook input, refresh the chart and confirm the visual updates. Complete the AI prompt/source log with tool, prompt, output, verification and owner.

Suggested timing

- 8 minutes - Validate data and define metrics
- 10 minutes - Build the Excel chart
- 15 minutes - Paste and format the native chart
- 8 minutes - Write insight, source and limitation
- 4 minutes - Refresh test and submit

Included inputs and assets

- Synthetic CSV and formula-enabled XLSX mock data
- Editable PowerPoint start and worked-example files
- PROMPTS.md with copy-ready data-analysis, infographic, Gemini Notebook and text-to-image prompts
- AI Prompt Lab worksheet for provenance and human-change records

Submission

Submit the editable PPTX, completed XLSX, source CSV, prompt/source log, used assets and checklist. Record every AI tool, prompt, source, rejected claim and material human change.

Debrief

The chart is native/editable, uses an appropriate form, retains source context and makes a proportionate product-promotion recommendation.